# Rent Analysis On and Off Campus at UC San Diego

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**Abstract**

This report evaluates whether UC San Diego on-campus housing prices are representative of nearby off-campus market rents. We construct a geospatial rental dataset for La Jolla Village and University City using SanGIS parcel and zoning data, RentCast rental estimates, and OpenStreetMap amenity locations. A log-linear regression model predicts rent per square foot from proximity to La Jolla Shores, UTC, and nearby parks. Applying the model to a Seventh College Tower West apartment suggests that the observed on-campus price of $7.51 per square foot per month is substantially higher than the model prediction of $3.74 and outside the 95% conformal prediction interval of $1.91 to $4.63. These results indicate that UCSD on-campus housing is not priced comparably to the surrounding off-campus rental market when evaluated by rent per square foot.

## Contents

**1 Background** **2**

**2 Dataset** **2**

**3 Model** **5**

**4 Model Implementation** **7**

**5 Limitations** **9**

**6 Conclusion** **9**

# 1 Background

One of the steepest costs for a student at UCSD is housing; students can expect to pay between $11,000 and $14,000 on their dorms for about a 9 month period. Between the already-high property values in San Diego and the heavy operating costs of HDH (UCSD's housing and dining department), we're interested in understanding the factors that play a role in rent cost in the area, and if UCSD's pricing is representative of the local property market. If it isn't representative, it's important to understand why that is and what that difference means for the students and families who foot the bill

To answer these questions, we built a regression model for rental rates based on property proximity to high-value amenities such as schools, parks, shopping centers, and the beach. Using this model, we predict what housing rates "should be" based on our null set of off-campus properties, allowing us to compare the predicted vs actual. Originally, we intended to perform some simple analysis on year-over-year increases in rent on and off campus. Unfortunately, due to the lack of publicly available data from HDH, we cannot conduct analysis on year-over-year increases and must stick with current rate analysis; HDH failed to provide additional information when contacted.

# 2 Dataset

Building our dataset required the integration of SanGIS geospatial data, RentCastAPI, and OSMnx (a Python library to interact with and download OpenStreetMap data). We first took San Diego parcel data and filtered for parcels within 5000 feet (about a mile) from the center of La Jolla Village and University City. These are two of the most popular neighborhoods for off-campus students, and the locations we wanted to utilize for our data. They are also very close to campus which should prevent location-based features from being biased towards off-campus data. Next, we spatially-joined these parcels with zoning data (also from SanGIS), filtering parcels for "Single-Family Detached", "Single Family Multiple-Units", and "Multi-Family Residential" homes. Housing near UCSD mostly consists of these zoning types, with only a few homes outside of this scope. At this point we have polygons of residential areas around our target communities, ready for rent price calculations:

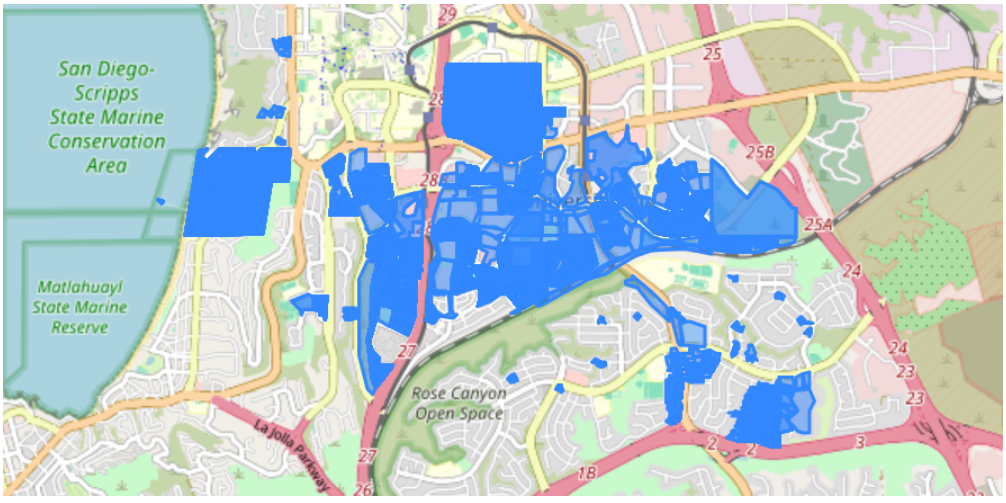



**Figure 1:** Residential parcel polygons near La Jolla Village and University City after zoning filters.

We decided to use RentCastAPI for our pricing data; this API can take geographical coordinates or addresses. Since addresses are not publicly displayed on SanGIS anymore, passing geographical coordinates is the more practical option. One query gives up to 50 results, each within a specified area around the queried coordinates. Because of this we implement a grid sample on the parcel polygons and query for prices within 0.25 miles of the coordinates (Note: our research indicates that using hexagons for the grid sample preserves actual distances better, but we anticipate that any differences are negligible. As it is, the grid points are spaced far-enough apart that search areas don't overlap). Our grid produces 145 points to be sampled:

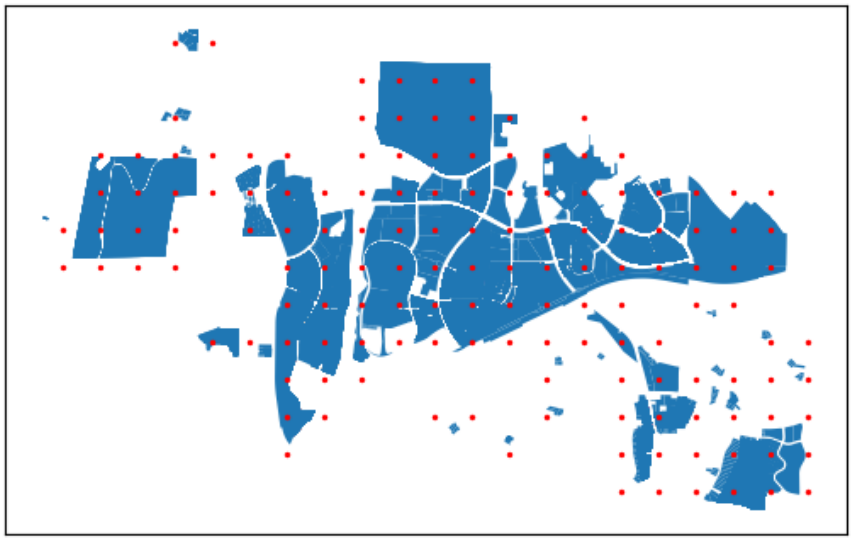


**Figure 2:** Grid sample used to query rental estimates around residential parcels.

Note: the grid's polygons were used in the spatial join instead of the centroids, leading to some centroids not landing in the zoning area. Since we pass on an area search to RentCast, this allows us to reach small "islands" of zoned residential land.

With the grid sample centroids and their geographic coordinates, we can call RentCast. RentCast provides 50 queries for free for each user, so 3 keys were enough to generate the entire dataset. We queried for median rental estimates and square footage, using rent per square foot as our target variable. This metric was very important to us: HDH has previously justified their housing pricing using price-per-bed. This is heavily biased in favor of on-campus, since a large off-campus master bedroom would count as one bed and an on-campus triple bedroom would count for 3 beds. For more information: `https://regents.universityofcalifornia.edu/regmeet/may22/f8attach.pdf`

The result of our RentCast queries is the following dataset, appearing bimodal with a concentration around $3-3.5 per square foot per month for rent. We removed all samples that didn't return rent data, resulting in a set of 133 neighborhoods and 4,924 properties. The only outlier in the dataset is a luxury neighborhood next to UTC. Even a neighborhood in La Jolla Shores within a quarter mile of the beach was only around $4-4.5 per square foot per month:

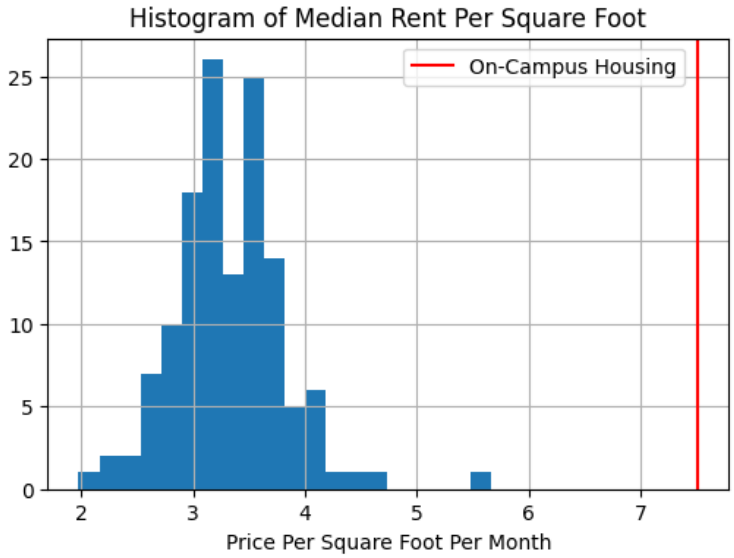



**Figure 3:** Distribution of off-campus rent estimates measured in dollars per square foot per month.

Our final step is to calculate distances to amenities for each property. While we wanted to use nearby shopping centers, UTC's proximity proved to overpower other shopping center metrics and was chosen instead as a feature. We also selected La Jolla Shores (the nearest beach in the area) and distance to downtown. Given the great distance from La Jolla to downtown San Diego, we anticipated that this might not be an effective feature since the percentage differences in distance would be quite similar.

These features were simple to add, since they have a central location and can be easily calculated in GeoPandas. However, our dataset required OpenStreetMap to calculate distances to the nearest school, park, and bus stop. From here we queried for our amenities and built a set of all schools, parks, and bus stops in the area. Using GeoPandas, we then could calculate the distance to the nearest of each amenity for each property.

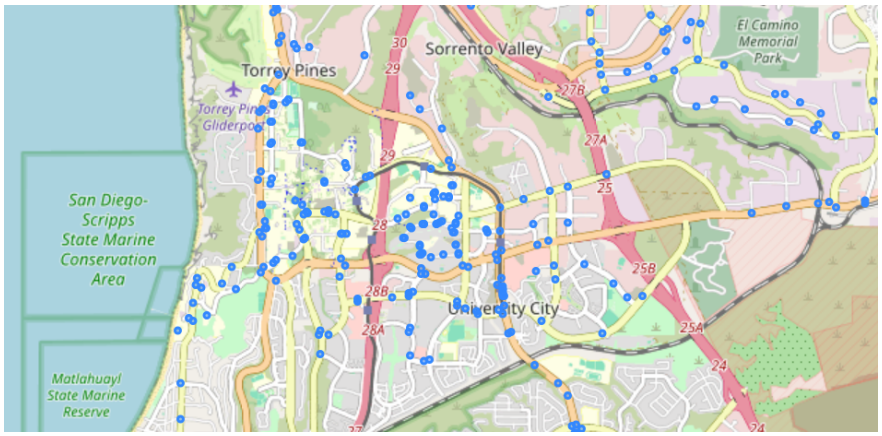



**Figure 4:** OpenStreetMap bus stop locations used as candidate amenity features.

# 3 Model

Finally, our dataset is built, and a model can be built using our features. Some issues were worth considering for our model; multicollinearity was expected to occur in some way, especially with the property data along the line between La Jolla Shores and UTC. Since these distances can depend on each other, we risk stability issues using both. In order to correct for this, we will check the Variance Inflation Factor (VIF) for each variable in the model we make. Another possible issue is our use of median rent price in each sample, which restricts the visibility of strong price outliers in some communities. Since it's still our best option for a typical-price metric, we keep it. We may also violate assumptions in our regression model. This will be considered after we construct our model, so we can test these assumptions using residuals or statistical tests.

To construct our model, we can brute-force different regression formulas (since the number of possible variables is only 6) and compare model criterion values for each. We test both an unscaled response variable and a log response variable (log median rent) given the success of log models in previous assignments. From here, each formula is fitted in statsmodels and both AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) are calculated to evaluate our models. Unsurprisingly, our log models had much better (lower) criterion values than the unscaled models, and we had two "best models": The lowest AIC came from "log median rent $\sim$ distance from La Jolla Shores + distance from UTC + distance from nearest park." The lowest BIC came from "log median rent $\sim$ distance from La Jolla Shores + distance from UTC." This difference makes sense because BIC has a greater penalty for the number of variables. The models are very similar and our number of variables is already small enough, so we choose the AIC model. With an R-squared of 0.281, the fit isn't great but still explains the data fairly well; our F-stat's p-value is 2.77e-09 to support the significance of our model. With coefficients, our model is:

$$y = 1.5498 - 1.394 \cdot 10^{-5}a - 2.275 \cdot 10^{-5}b - 3.188 \cdot 10^{-5}c \tag{1}$$

where y is median rent per square foot, a is the distance to La Jolla Shores in feet, b is the distance to UTC in feet, and c is the distance to the nearest park in feet.

The coefficients of this model make sense. The intercept indicates that if your property was right next to La Jolla Shores, UTC, and a park, then the expected rent per square foot would be $e^{1.5498} = 4.71$ dollars. The other coefficients are negative since the farther you are from desired amenities, the lower your expected rent will be. For a coefficient $e^{\beta}$ of the covariate coefficients, the expected rent per square foot increases $e^{\beta}$ for an increase of one in the covariate, holding all others constant. Note: The coefficients are very small because we used units of feet, which comes from the local projected coordinate reference system we used.

Next, we check for multicollinearity. As it turns out, our calculated variance inflation coefficients were less than 2 for all coefficients in both models. It seems that our set of distance-based coefficients was small enough to prevent collinearity. As we move through our regression model assumptions, we also plot residuals vs fitted values to identify any patterns in the residuals. Visual randomness in our residuals is a good indicator of a proper model for the relationship:

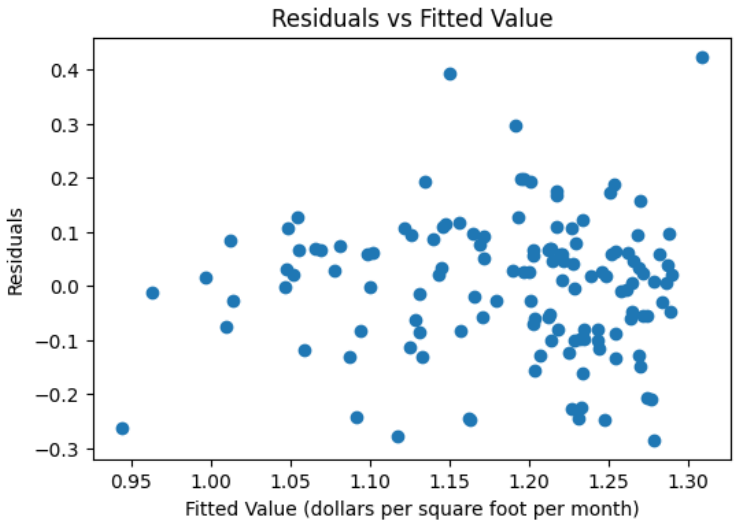



**Figure 5:** Residuals versus fitted values for the selected log-linear regression model.

Upon inspection, the residual plot appears to be random. Although there may seem to be greater variability at larger fitted values, this is likely due to the higher concentration of observations on the right-hand side of the plot. Based on this, it's reasonable to assume that the relationship between our independent variables and our response variable can be appropriately modeled by linear regression; there also appears to be homoscedasticity.

Given random residuals with (assumed) constant variance, we move onto normality of variance. When using a QQ Plot to compare to a normal distribution, our data fits extremely well, with some minor fitting issues for a few outliers:

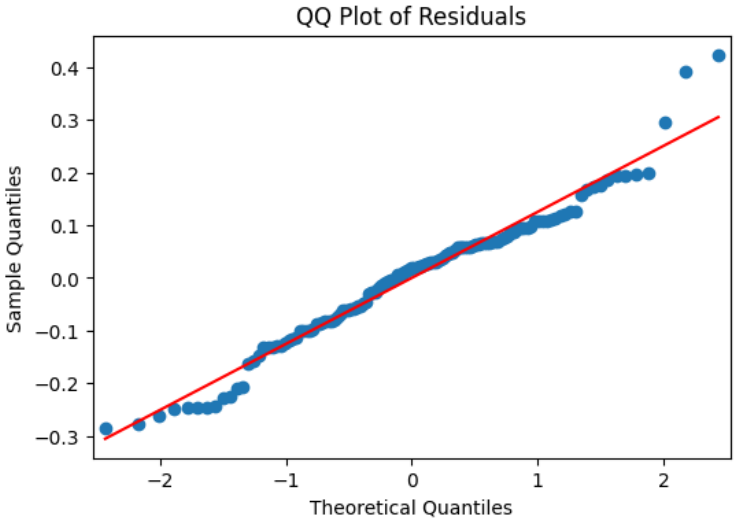



**Figure 6:** Q-Q plot for model residuals.

Our sample data appears to be independent when using the Autocorrelation Function:

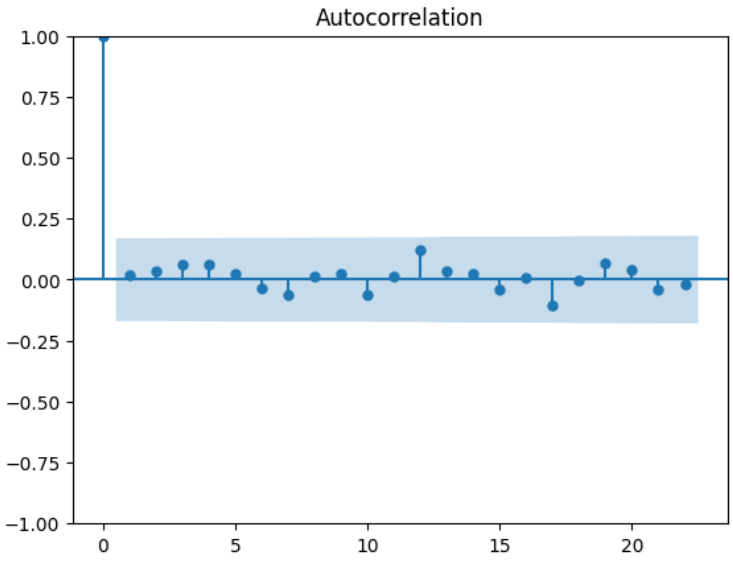



**Figure 7:** Autocorrelation function for residuals.

So far, we've shown independence of our data, randomness of residuals, a linear relationship, normally-distributed and homoscedastic variance, and a lack of multicollinearity. These are the main assumptions of the linear model, supporting its use in our analysis. From here, we can fit the model to the features of an on-campus dorm at UCSD and test whether it's representative of a typical rental property in the local area.

# 4 Model Implementation

To test the fit of our model on UCSD housing, we will use a Seventh College Tower West apartment as it has several attractive features: Its apartment-style layout allows us to properly consider the square footage of the entire apartment; with a typical dorm we'd have to account for common areas which would be difficult to accomplish. The floor plans of Seventh College also feature the exact square footage, which is only available for family apartments otherwise. Using our AIC model, we must collect distance from La Jolla Shores, distance to UTC, distance from the nearest park, and the square footage for the response variable. Tower West is about 12,038 feet from La Jolla Shores, about 10,616 feet from UTC, and about 436 feet from the nearest park. The square footage of a typical apartment at Tower West is 411 square feet.

Before fitting the model to this data, some assumptions were made concerning the data. While utilities are covered while living on-campus, separate fees exist for laundry, gym use, etc. Because these amenities vary from property to property, we ignore them in our calculations. From brief research, utilities may be fairly negligible in our calculations anyway; they're often only about 50 dollars per person. If the gap in fitted vs actual rent pricing is large enough, we can be sure that accounting for utilities wouldn't change our conclusion.

We also have to add individual rent per person for on-campus housing, and the number of people per dorm can vary. Less people in an apartment means cheaper rent per square foot (even considering higher rent for singles), so we pick a two-single apartment as the most conservative pricing: an unusual arrangement. If their rent doesn't fit the model, more-crowded apartments

also won't fit by extension. A 9 month contract for each single totals $13,889, or $1543.22 a month. Between two people, this totals $3086.44 for 411 square feet or $7.51 per square foot per month. Superimposing this price onto the off-campus distribution in red:

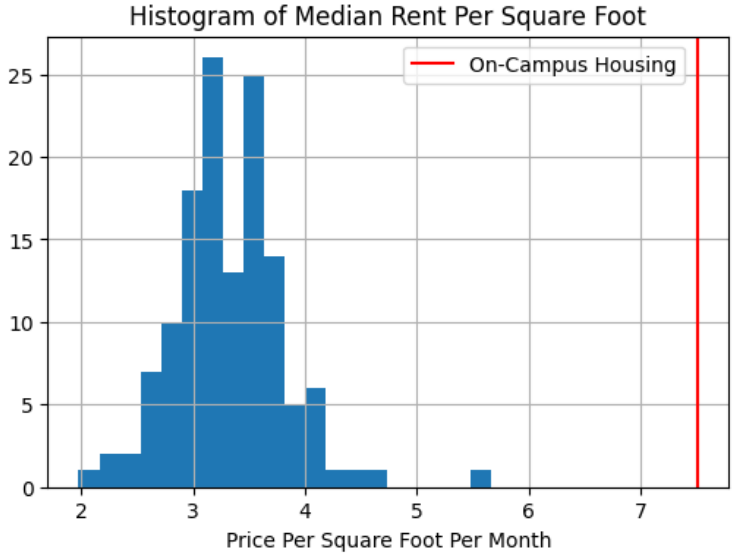



**Figure 8:** Off-campus rent distribution with the Seventh College Tower West price marked in red.

This rent price clearly doesn't belong to the null distribution, over double the price of a typical unit in terms of square footage. When fitting the features of Tower West to our model, we get a predicted price of $3.74 per square foot, half of the actual price. Visually our conclusion is obvious, but we can also use conformal inference to test this. Under $\alpha = 0.05$, we obtain an interval of (0.647, 1.533), meaning that we expect 95% of new points with Tower West's features to fall within this interval for log median rent price, or actual median rent price to fall within (1.91, 4.63) dollars.

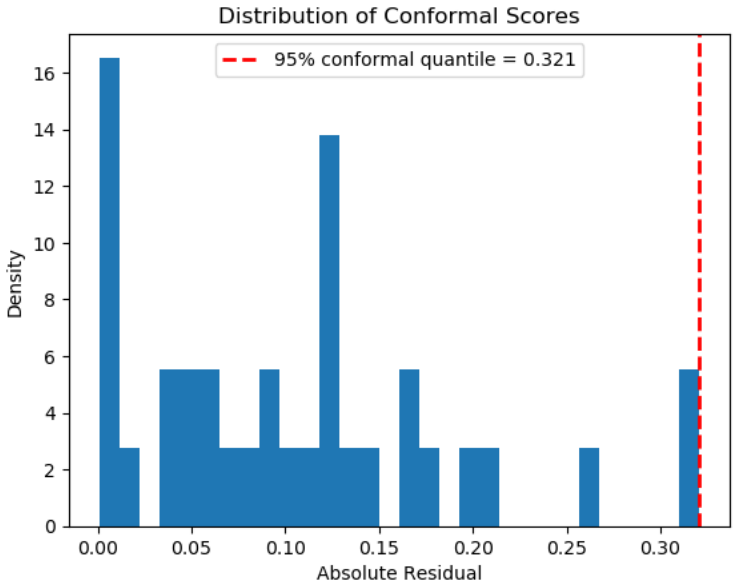



**Figure 9:** Conformal prediction interval for a property with Seventh College Tower West features.

This makes the calculated $7.51 per square foot inconsistent with the modeled off-campus distribution. Therefore, we conclude that on-campus housing is not representative of fair market value for rent and is disproportionately high when evaluated by rent per square foot. While there may exist valid reasons for HDH to operate on-campus housing at these rates (high mortgage on housing properties, custodial staff expenses, etc.), their claim to offer rates "20% below market value" are unfounded based on our analysis. These claims likely stem from their use of "per bed" pricing of rental properties; this allows HDH to split their room rent among up to three people, while all off-campus rooms remain as one-person rooms regardless of their size. We believe our pricing system is a less biased approach and more accurately communicates the value of a property.

# 5 Limitations

There are some important limitations and shortcomings in our analysis. First, our selected regression model had an $R^2$ value of 0.281, meaning that it explained only about 28% of the variation in rent per square foot. This suggests that location and distance to amenities matter, but they are not the whole story. Other variables could also affect rent, including building age, unit quality, safety, furniture, lease terms, parking, included utilities, and building amenities.

Second, we only tested one UCSD housing example. We chose Seventh College Tower West because it has an apartment-style layout and publicly available square footage, which made it possible to calculate rent per square foot. However, one example is not enough to describe all UCSD housing as each residence was built at different time periods. Other colleges, room types, apartment layouts, and occupancy arrangements may produce different comparisons.

Third, we did not have detailed historical HDH pricing data. Because of this, we could not study whether UCSD housing prices increased faster or slower than nearby off-campus rents over time. This limited our analysis to a current-rate comparison rather than a trend analysis.

Fourth, we did not fully account for utilities, laundry, gym access, custodial services, maintenance, staffing, construction costs, or other fees and benefits included in or associated with on-campus housing. These factors could affect the comparison between UCSD housing and off-campus rentals. However, the gap between the actual Seventh College Tower West price of $7.51 per square foot and the model prediction of $3.74 per square foot is large enough that these omitted factors would need to be substantial to fully explain the difference.

# 6 Conclusion

This project found that nearby off-campus rent was usually around $3 to $3.50 per square foot per month. Our model predicted that the Seventh College Tower West example should cost about $3.74 per square foot, while the actual UCSD cost was about $7.51 per square foot. The main takeaway is that UCSD housing looks much more expensive when measured by rent per square foot, even though it may look more affordable under a price-per-bed comparison. We are not claiming that UCSD has no additional costs or benefits, but based on the data we had, the claim that this housing is 20% below market rate was not supported by our rent-per-square-foot analysis. For the Seventh College example we tested, UCSD housing was not representative of nearby market rent when measured by rent per square foot.